

Score  $\geq 9$  is classified as SSc.

Patients with suspected early SSc are referred to centres that are experienced in SSc diagnosis and care.

# Diagnostic procedures

## Antinuclear antibodies

Autoantibodies targeting characteristic nuclear antigens are one of the hallmarks of systemic sclerosis (SSc). The frequency of antinuclear antibodies (ANA) in SSc patients approaches 95%, consistent with literature reports of 85% to 99%. In one study, 86.6% of ANA-positive patients had SSc-specific antibodies, with 96.4% directed against five major antigens: **ACA (anti-centromere)**, **anti-topoisomerase I (anti-Scl-70)**, **anti-RNA polymerase III**, **anti-PM/Scl**, and **anti-U1-RNP**.

These SSc-specific autoantibodies are largely mutually exclusive; co-occurrence in the same patient is rare. Their presence is strongly associated with distinct clinical phenotypes and organ involvement, supporting their use in an outcome-based classification system for SSc that aligns with current clinical stratification practices.

For a more detailed description of autoantibodies linked to overlap syndromes, please see Section - Systemic Sclerosis Overlap Syndromes.

## Capillaroscopy

Capillaroscopy—performed using videocapillaroscopy, stereomicroscopy, or dermatoscopy—is a well-established, non-invasive technique for evaluating nailfold capillary abnormalities. It plays a key role in differentiating **primary**

**Raynaud's phenomenon** from **secondary Raynaud's phenomenon** due to SSc.

In SSc, characteristic microvascular changes include capillary enlargement, microhaemorrhages, avascular areas, and abnormal capillary architecture (e.g., bushy or ramified capillaries). These findings support early diagnosis, even before full clinical manifestations of SSc develop.

Various scoring systems have been proposed to standardize interpretation. Automated analysis platforms are under development and may facilitate broader clinical application in the future.

## Organ involvement and diagnostic work-up

### *Raynaud's phenomenon*

Raynaud's phenomenon is defined by episodic vasospasm of the digital arteries and arterioles, leading to a classic triphasic colour change: **blanching (pallor)**, **cyanosis (blue)**, followed by **reactive hyperaemia (red)**.

It affects over 90% of SSc patients and most commonly involves the fingers, though toes, nose, ears, and tongue can also be affected. Typical triggers include **cold exposure** and **emotional stress**.

- **Primary Raynaud's phenomenon** arises from functional vascular dysregulation without underlying connective tissue disease.
- **Secondary Raynaud's phenomenon**, as seen in SSc, involves both functional impairment and **structural vascular damage** (e.g., intimal proliferation, vessel wall thickening), increasing the risk of **digital ulcers** and tissue loss.

Differentiation between primary and secondary forms requires: - **Nailfold capillaroscopy** - **Autoantibody testing (ANA and SSc-specific antibodies)**

Abnormal capillaroscopy and positive SSc-specific autoantibodies strongly suggest an underlying connective tissue disorder such as SSc.

*Autoantibodies in systemic sclerosis*

| Antibodies                  | Organ involvement                      |
|-----------------------------|----------------------------------------|
| SSc-specific autoantibodies |                                        |
|                             | Centromere (ACA)                       |
|                             | Topoisomerase-1 (Scl-70)               |
|                             | RNA polymerase III                     |
|                             | PM/Scl                                 |
| U1-RNP                      | Joint involvement                      |
| SSc-associated antibodies   |                                        |
| Ro, La                      | Parotid involvement (Sjögren syndrome) |
| CCP                         | Rheumatoid arthritis                   |
| Rheumatoid factor           | Rheumatoid arthritis                   |
| Mitochondrial (M2)          | Liver (primary biliary cholangitis)    |

**Abbreviations:** CCP – cyclic citrullinated peptide; SSc – systemic sclerosis.